

Assignment2

1. $X \rightarrow a$
 $X \rightarrow Y$
 $Z \rightarrow d$
 $Z \rightarrow XYZ$
 $Y \rightarrow c$
 $Y \rightarrow \epsilon$

Find out the FIRST(X), FIRST(Y), FIRST(Z)

Find out the FOLLOW(X), FOLLOW(Y), FOLLOW(Z)

Construct the parsing table using FIRST and FOLLOW

2. $S \rightarrow AaAb$
 $S \rightarrow Bb$
 $A \rightarrow \epsilon$
 $B \rightarrow \epsilon$

Find out the FIRST(S), FIRST(A), FIRST(B)

Find out the FOLLOW(S), FOLLOW(A), FOLLOW(B)

Construct the parsing table using FIRST and FOLLOW

3. $C \rightarrow PF \text{ class id } XY$
 $P \rightarrow \text{public} \mid \epsilon$
 $F \rightarrow \text{final} \mid \epsilon$
 $X \rightarrow \text{extends id} \mid \epsilon$
 $Y \rightarrow \text{implements I} \mid \epsilon$
 $I \rightarrow \text{id } J$
 $J \rightarrow I \mid \epsilon$

Find out the FIRST(C), FIRST(P), FIRST(F), FIRST(X), FIRST(Y), FIRST(I), FIRST(J)

Find out the FOLLOW(C), FOLLOW(P), FOLLOW(F), FOLLOW(Y), FOLLOW(I), FOLLOW(J)

Construct the parsing table using FIRST and FOLLOW

4. $S \rightarrow Xa$
 $X \rightarrow YZ$
 $Y \rightarrow B \mid \epsilon$
 $Z \rightarrow c \mid \epsilon$

Find out the FIRST(S), FIRST(X), FIRST(Y), FIRST(Z)

Find out the FOLLOW(S), FOLLOW(X), FOLLOW(Y), FOLLOW(Z)

Construct the parsing table using FIRST and FOLLOW